Software Engineering (Sessional) Final Report

Course: CSE-434



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Executive Summary

GUIDE (delete before submission)

Write this LAST, after all other chapters are complete. Length: **one page (250–400 words)**. Cover, in this order: (1) the problem, (2) your proposed solution and main features, (3) the process model and technologies used, (4) the AI component (if any), (5) key results from testing/deployment, and (6) the main conclusion/future work. Do **not** include citations, figures or tables here.

[TODO: Write the executive summary here.]

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Chapter 1

Introduction

1.1 Goals and Objectives of the Project

GUIDE (delete before submission)

List the main **goal** (1–2 sentences) and 3–6 **specific, measurable** objectives (SMART). Example: "Reduce manual attendance recording time from 10 min to under 1 min per class."

[TODO: Write goals and objectives.]

1. **[TODO: Objective 1 (measurable)]**
2. **[TODO: Objective 2 (measurable)]**

1.2 Scope of the Work

GUIDE (delete before submission)

State clearly what the system **will** do (in scope) and what it **will not** do (out of scope).

1.2.1 Current Situation and Context

GUIDE (delete before submission)

Describe the existing problem or manual system and why a new system is needed. Use facts, statistics or interviews/surveys with citations.

[TODO: Describe the current situation.]

1.2.2 Competing Products (Available in Market)

GUIDE (delete before submission)

Compare 3–5 existing products. Summarise features and *limitations* in a table, then explain how your system differs.

Table 1.1: Comparison with existing products

Product	Key Features	Limitations	Our Advantage
Product A	[TODO: ...]	[TODO: ...]	[TODO: ...]
Product B	[TODO: ...]	[TODO: ...]	[TODO: ...]

1.3 System Overview

GUIDE (delete before submission)

High-level description: what the system does, main features, target users, and (if applicable) the AI capability. Include one overview figure.

[TODO: Write system overview.]

[Figure placeholder: `system_overview.png`]

Figure 1.1: System overview

1.4 Structure of the Document

GUIDE (delete before submission)

One or two sentences per chapter describing its contents. Update after finishing the report.

The remainder of this report is organised as follows. Chapter 2 presents the project management plan. Chapter 3 specifies the requirements. Chapter 4 describes the architecture. Chapter 5 details the design. Chapter 6 explains the implementation. Chapter 7 reports testing and quality assurance. Chapter 8 covers deployment and go-live, and Chapter 9 presents the sustainability plan.

1.5 Terms, Acronyms, and Abbreviations Used

GUIDE (delete before submission)

Define every technical term, acronym and abbreviation used in the report (alphabetical order).

Table 1.2: Terms, acronyms and abbreviations

Term	Definition
API	Application Programming Interface
CI/CD	Continuous Integration / Continuous Deployment
ER	Entity–Relationship
UAT	User Acceptance Testing
VAPT	Vulnerability Assessment and Penetration Testing

Chapter 2

Project Management Plan

2.1 Project Organization

GUIDE (delete before submission)

Describe the team structure (roles such as project manager, analyst, developer, tester) and communication channels / meeting frequency.

[TODO: Describe the team organisation.]

2.1.1 Individual Contribution to the Project

GUIDE (delete before submission)

Detail what EACH member contributed. Use the tick table and add 2–3 sentences per member. Add rows for more members.

Individual contributions are shown in Table 2.1.

Table 2.1: Individual contribution to the project

Member	Req. Spec.	Planning	Design	Model Building	User Interface	Testing	Deployment	Docs
Team Member 1 Name	✓							
Team Member 2 Name		✓						
Team Member 3 Name			✓					

2.2 Process Model Used

GUIDE (delete before submission)

Name the lifecycle model (Waterfall, Incremental, Scrum/Agile, Spiral, Prototyping...) and show a diagram of how you applied it (sprints, iterations, deliverables).

[TODO: Describe the process model.] Example citation: [1, 2].

2.2.1 Rationale for Choosing Lifecycle Model

GUIDE (delete before submission)

Justify the choice using project size, requirement stability, team size, risk, and customer involvement. Mention why alternatives were rejected.

[TODO: Justify the lifecycle model.]

2.3 Risk Analysis

GUIDE (delete before submission)

Identify risks (technical, schedule, resource, AI/data-related, security) with probability, impact, and mitigation. Use the risk-exposure = probability × impact approach.

Table 2.2: Risk analysis

ID	Risk	Prob.	Impact	Mitigation
R1	Team member unavailable	Med	High	Cross-training; shared task board
R2	Insufficient / biased training data (AI)	High	High	Data augmentation; bias audit
R3	**[TODO: ...]**			

2.4 Constraints to Project Implementation

GUIDE (delete before submission)

Mention schedule, budget, software, hardware, data, legal and skill constraints.

[TODO: List constraints.]

2.5 Tools/Languages Requirements

2.5.1 Hardware Requirements

GUIDE (delete before submission)

Development and deployment hardware (CPU/GPU, RAM, storage, devices). Table recommended.

[TODO: List hardware requirements.]

2.5.2 Software Requirements

GUIDE (delete before submission)

IDEs, languages, frameworks, libraries (with versions), databases, DevOps/design/collaboration tools.

[TODO: List software requirements.]

2.6 Project Timeline and Schedule

GUIDE (delete before submission)

Provide a Gantt chart with phases and milestones. Edit the example below (package `pgfgantt`).

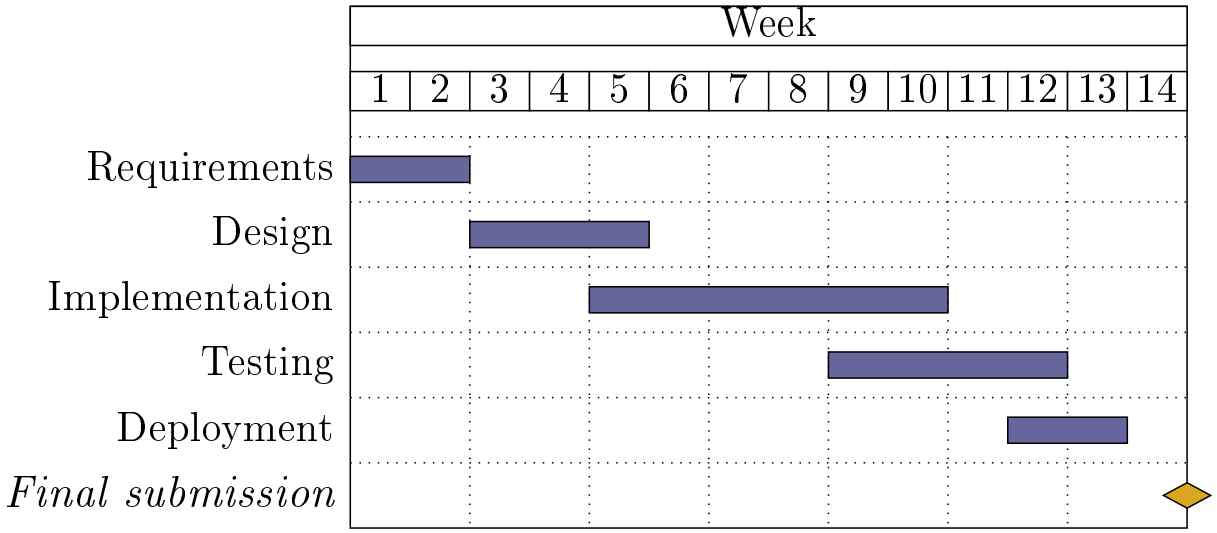


Figure 2.1: Project timeline (Gantt chart)

2.7 Estimated Budget

GUIDE (delete before submission)

Break down estimated costs (hosting, domain, cloud/GPU, tools, devices, API usage, human effort). Give totals in BDT (and USD if relevant).

Table 2.3: Estimated budget

Item	Unit	Qty	Cost (BDT)
Cloud hosting	month	6	[TODO: ...]
Domain name	year	1	[TODO: ...]
Total			[TODO: ...]

2.8 Technological, Societal, & Environmental Impact of the Project

GUIDE (delete before submission)

Discuss positive AND negative impacts on individuals, society, culture, economy and the environment (e.g. energy use of AI training/hosting, e-waste, accessibility, job impact, ethics).

[TODO: Discuss impacts.]

Chapter 3

Requirement Specifications

3.1 Stakeholders for the System

GUIDE (delete before submission)

List all stakeholders (users, admins, client, developers, regulators...) with their role and interest/influence.

Table 3.1: Stakeholders

Stakeholder	Role	Interest in the system
End user	[TODO: ...]	[TODO: ...]
Administrator	[TODO: ...]	[TODO: ...]

3.2 Use Case Diagram with Graphical and Textual Description

GUIDE (delete before submission)

Give the use case diagram (UML) and, for **each** use case, a textual description table: ID, name, actors, preconditions, postconditions, main flow (steps), alternative/exception flows. Tools: draw.io, StarUML, PlantUML, Lucidchart.

[Figure placeholder: `usecase_diagram.png`]

Figure 3.1: Use case diagram

Table 3.2: Textual description of use case UC-01

Use Case ID	UC-01
Name	**[TODO: e.g. Upload Image]**
Actor(s)	**[TODO: Primary / secondary actors]**
Precondition	**[TODO: ...]**
Postcondition	**[TODO: ...]**
Main flow	1. **[TODO: ...]** 2. **[TODO: ...]** 3. **[TODO: ...]**
Alternative flow	**[TODO: ...]**
Exception flow	**[TODO: ...]**

3.3 Activity Diagram

GUIDE (delete before submission)

Show the workflow of major processes (swimlanes recommended). One diagram per major use case/process.

[Figure placeholder: `activity_diagram.png`]

Figure 3.2: Activity diagram of **[TODO: process]**

3.4 Static Model – Class Diagram

GUIDE (delete before submission)

Show classes, attributes, methods, visibility and relationships (association, aggregation, composition, inheritance) with multiplicities.

3.5 Dynamic Model – Sequence Diagram

GUIDE (delete before submission)

Show object interactions for the key scenarios (one diagram per important use case). Include actor, UI, controller, services, DB and AI/external API lifelines.

[Figure placeholder: `class_diagram.png`]

Figure 3.3: Class diagram

[Figure placeholder: `sequence_diagram.png`]

Figure 3.4: Sequence diagram of **[TODO: scenario]**

3.6 Safety and Security Requirements

3.6.1 Access Requirements

GUIDE (delete before submission)

Who has authorised access to functionality and data, under what circumstances, and to which parts. Include authentication, authorisation, roles (RBAC table).

[TODO: Write access requirements.] See also [3].

3.6.2 Integrity Requirements

GUIDE (delete before submission)

Required integrity of databases, files and the product itself: data accuracy, validation, consistency, backups, audit logs, transaction handling.

[TODO: Write integrity requirements.]

3.6.3 Privacy Requirements

GUIDE (delete before submission)

What the product must do to protect personal data and comply with privacy laws (e.g. Bangladesh Data Protection Act / GDPR where applicable): consent, minimisation, encryption, retention, anonymisation.

[TODO: Write privacy requirements.]

3.7 AI-integration Requirements

GUIDE (delete before submission)

Specify requirements for the AI component: functional (inputs/outputs, tasks), data (sources, quality, labelling), performance (accuracy, precision/recall, latency), ethical (fairness, transparency, explainability, human oversight), robustness, and monitoring/retraining. If your project has no AI part, write a short justification and (with your teacher's approval) mark this section "Not applicable".

Table 3.3: AI-integration requirements

ID	Requirement	Must/Want	Metric
R-AI-01	Model shall classify input with accuracy $\geq$ **[TODO: X]**%	Must	Accuracy
R-AI-02	Inference latency shall be below **[TODO: X]** s	Want	Latency

Chapter 4

Architecture

4.1 Architectural Model/Style Used

GUIDE (delete before submission)

Name the style (Layered, MVC, Client–Server, Microservices, Event-driven, Pipe-and-filter...).

[TODO: Describe the architectural style.]

4.1.1 Rationale for Choosing the Architectural Model/Style

GUIDE (delete before submission)

Explain why the style fits your quality attributes (scalability, maintainability, performance, security) and why alternatives were not chosen.

[TODO: Justify the architectural style.]

4.2 Architectural Design (High-Level Design)

GUIDE (delete before submission)

Overall architecture diagram: layers, components, external systems and their interactions, with a short description of every component.

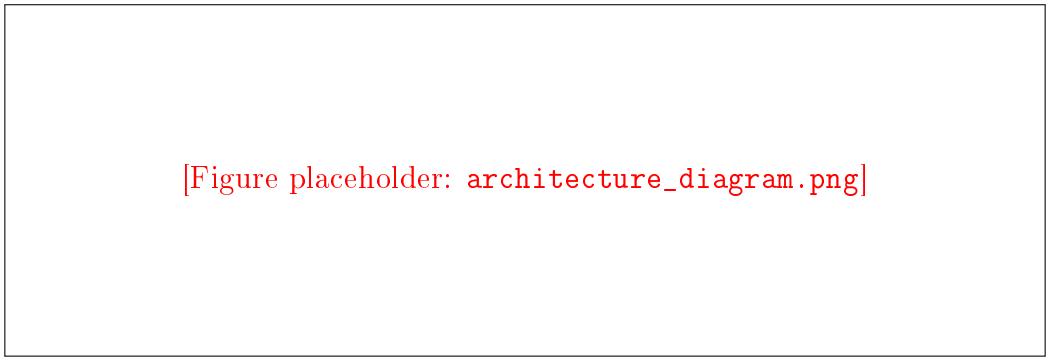


Figure 4.1: High-level system architecture

4.3 Technology, Software, and Hardware Used

GUIDE (delete before submission)

Detailed tech-stack table WITH justification for each choice.

Table 4.1: Technology stack and justification

Layer	Technology	Justification
Frontend	**[TODO: React / Flutter...]**	**[TODO: ...]**
Backend	**[TODO: Django / Node.js...]**	**[TODO: ...]**
Database	**[TODO: PostgreSQL...]**	**[TODO: ...]**
AI/ML	**[TODO: PyTorch / TensorFlow...]**	**[TODO: ...]**
Hosting	**[TODO: AWS / Render...]**	**[TODO: ...]**

4.4 AI-integration Architecture

GUIDE (delete before submission)

Show how the AI model is integrated: data pipeline, training pipeline, model registry/storage, inference service (REST/gRPC/embedded), API calls to external LLMs, monitoring, and fallback behaviour.

[Figure placeholder: `ai_architecture.png`]

Figure 4.2: AI-integration architecture

4.5 CI/CD Pipeline

GUIDE (delete before submission)

Describe your CI/CD tools (GitHub Actions, GitLab CI, Jenkins...) and pipeline stages: lint → build → unit test → integration test → security scan → package → deploy. Include a pipeline diagram and a snippet of the workflow file.

```yaml
name: CI
on: [push, pull_request]
jobs:
  build-test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-python@v5
        with: { python-version: "3.11" }
      - run: pip install -r requirements.txt
      - run: pytest --cov=app
```

Listing 4.1: Example GitHub Actions workflow (`.github/workflows/ci.yml`)

4.6 Git Management (Version Controlling)

GUIDE (delete before submission)

Branching strategy (e.g. Git Flow / trunk-based), branch naming, commit message convention, pull-request and code-review rules, protected branches, tagging/releases, and a screenshot of the commit graph/contribution statistics. Give the repository URL.

[TODO: Describe the Git workflow.]

Chapter 5

Design

5.1 High-Level Design (HLD) [Component Level Design]

GUIDE (delete before submission)

Major components, their responsibilities and interfaces; design patterns used (Factory, Observer, Singleton, Strategy, Repository…) with the reason and a small UML snippet for each.

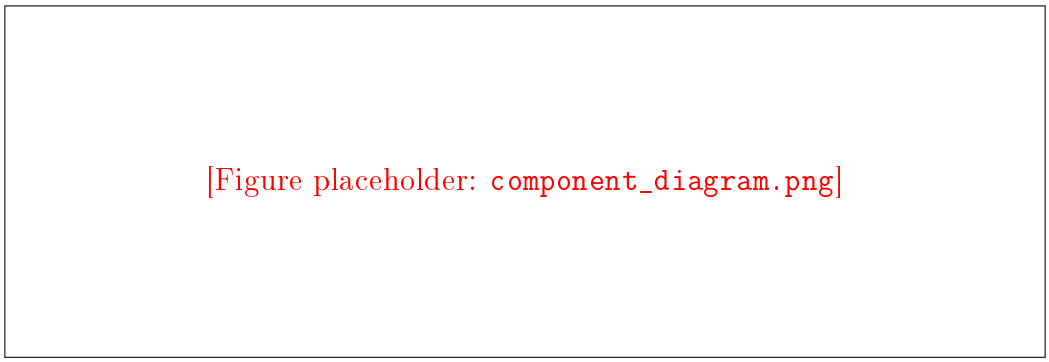


Figure 5.1: Component diagram

5.2 Low-Level Design (LLD)

GUIDE (delete before submission)

Detailed design of individual modules: algorithms, data structures, pseudocode, state diagrams.

Algorithm 1 Example: input validation algorithm

Require: uploaded file f
Ensure: VALID or INVALID
1: **if** extension(f) $\notin$ {jpg, png} **then**
2: **return** INVALID
3: **end if**
4: **return** VALID

5.3 Database Schema (ER Diagram & Schema)

GUIDE (delete before submission)

ER diagram, table structures (with data types), primary/foreign keys, relationships, indexes and normalisation (show up to 3NF/BCNF).

[Figure placeholder: `er_diagram.png`]

Figure 5.2: Entity–Relationship diagram

Table 5.1: Schema of table `users`

Column	Type	Key	Description
id	INT	PK	Unique user id
email	VARCHAR(100)	UNIQUE	Login e-mail

5.4 GUI (Graphical User Interface) Design

GUIDE (delete before submission)

Wireframes (Figma, Balsamiq), mockups and final screens with explanations of navigation and usability decisions.

[Figure placeholder: `gui_wireframe.png`]

Figure 5.3: Wireframe of the main screen

5.5 API Documentation

GUIDE (delete before submission)

List ALL APIs: endpoint, method, parameters, request/response examples, status codes, authentication. Export from Swagger/OpenAPI or Postman if possible.

Table 5.2: API summary

Method	Endpoint	Description	Auth
POST	`/api/v1/login`	User login	No
GET	`/api/v1/items`	List items	Bearer token

```
POST /api/v1/login
{ "email": "user@example.com", "password": "********" }

200 OK
{ "token": "eyJhbGciOi...", "expires_in": 3600 }
```

Listing 5.1: Sample request/response (JSON)

5.6 AI Model Design & Integration

GUIDE (delete before submission)

Model choice and justification, dataset (source, size, split, pre-processing), architecture/hyper-parameters, training setup, evaluation metrics and results, how the model is exposed to the application (API/embedded), prompt design for LLMs, and handling of failures/uncertainty.

[TODO: Describe the AI model design.]

5.7 Test Plan

5.7.1 Test Strategy and Approaches

GUIDE (delete before submission)

Levels of testing (unit, integration, system, acceptance), test types (functional, performance, security, AI-specific), tools, environments, entry/exit criteria, roles and schedule.

[TODO: Describe the test strategy.]

5.7.2 Test Case Design/Preparation (High-Level or Sample Test Cases)

GUIDE (delete before submission)

Give sample/high-level test cases here. Full detailed cases go to Chapter 7 and Appendix B.

[TODO: Provide sample test cases.]

Chapter 6

Implementation / Construction

6.1 Development Environment and Tools

GUIDE (delete before submission)

IDE, version control, local setup, server/cloud setup, containers (Docker), package managers.

[TODO: Describe the development environment.]

6.2 Key Implementation Details / Modules Developed

GUIDE (delete before submission)

For each important module: purpose, how it was implemented, and a SHORT annotated code excerpt (10–25 lines). Put full code in Appendix A / the repository.

```python
ALLOWED = {".jpg", ".jpeg", ".png"}

def is_valid_image(filename: str) -> bool:
    """Return True only for supported image extensions."""
    ext = filename[filename.rfind("."):].lower()
    return ext in ALLOWED
```

Listing 6.1: Example: image-extension validation

Listing 6.1 shows **[TODO: explain the code]**.

6.3 AI-integration Implementation

GUIDE (delete before submission)

How the AI model was implemented and wired into the system: data pre-processing code, training/fine-tuning scripts, model loading, inference endpoint, prompt templates, caching, error handling, and cost/latency considerations.

[TODO: Describe the AI implementation.]

6.4 Code Structure and Standards Followed

GUIDE (delete before submission)

Folder structure, naming conventions, coding standards (PEP 8, ESLint, Google Style...), linting/formatting tools, documentation/comments policy, code-review checklist.

```
project/
|-- backend/    (APIs, services, models)
|-- frontend/   (UI components)
|-- ml/         (datasets, notebooks, training scripts)
|-- tests/      (unit & integration tests)
|-- docs/
`-- .github/workflows/
```

Listing 6.2: Project folder structure

Chapter 7

Testing and Quality Assurance

7.1 Requirements/Specifications-Based System Level Test (Detailed with Test Results)

GUIDE (delete before submission)

First list the requirements (ID, statement, Must/Want), then give one test-case table per module, with ACTUAL results and Pass/Fail status. Copy the module table block for each module.

Table 7.1: Requirements/specifications-based requirement list

Requirement ID	Requirement Statement	Must/Want	Comment
R-Up-01	User can't upload image with invalid extension	Must	N/A
R-Up-02	Show error message if user uploads image with invalid extension	Want	N/A
R-Gen-01	User can't generate caption without uploading image	Must	N/A

Table 7.2: System level test cases – module: **[TODO: Upload]**

Project Name	Title of the Project	**Module Name**	**[TODO: Upload]**
Created By	**[TODO: Name]**	**Reviewed By**	**[TODO: Name]**
Date of Creation	**[TODO: dd-mm-yy]**	**Date of Review**	**[TODO: dd-mm-yy]**

TC ID	Scenario	Steps	Test Data	Expected	Actual	Status
TC-UP-01	Upload with invalid extension	1. Open UI 2. Choose file 3. Upload	new.pdf	Return to initial state	Returned to initial state	Passed
TC-UP-02	**[TODO: ...]**	**[TODO: ...]**	**[TODO: ...]**	**[TODO: ...]**	**[TODO: ...]**	**[TODO: ...]**

7.2 Traceability of Test Cases to Use Cases

GUIDE (delete before submission)

Traceability matrix: test cases vs. requirements / use cases. Every requirement must be covered by at least one test case.

Table 7.3: Traceability matrix

Test Case	R-Up-01	R-Up-02	R-Gen-01	UC-01	UC-02
TC-UP-01	✓			✓	
TC-UP-02		✓		✓	
TC-GEN-01			✓		✓

7.3 Techniques Used for Test Generation

GUIDE (delete before submission)

Black-box (equivalence partitioning, boundary value analysis, decision table, state transition), white-box (statement/branch/path coverage), experience-based, and AI-assisted/ automated generation. Say where each was applied.

[TODO: Describe test generation techniques.]

7.4 Unit Testing, Integration Testing

GUIDE (delete before submission)

Tools (pytest, JUnit, Jest. . .), number of tests, coverage report, results, and a screenshot of the test run.

[TODO: Report unit and integration testing.]

7.5 System Testing (UAT, PT, VT)

GUIDE (delete before submission)

UAT: users, tasks, feedback/SUS score. **Performance Testing**: tool (JMeter, Locust), response time, throughput. **Volume Testing**: behaviour with large data. Present results in tables/graphs.

[TODO: Report UAT, performance and volume tests.]

7.6 Security Testing (VAPT)

GUIDE (delete before submission)

Vulnerability assessment and penetration test findings (OWASP Top 10: SQL injection, XSS, broken authentication...). Tools: OWASP ZAP, Burp Suite, Nmap. Report findings, severity, and fixes.

Table 7.4: VAPT findings

ID	Vulnerability	Severity	Fix / Status
V-01	[TODO: ...]	[TODO: High/Med/Low]	[TODO: ...]

7.7 AI-specific Testing

GUIDE (delete before submission)

Model evaluation (accuracy, precision, recall, F1, BLEU/ROUGE...), confusion matrix, robustness/adversarial inputs, bias & fairness checks, hallucination checks for LLMs, drift monitoring, explainability checks, edge cases.

[TODO: Report AI-specific testing.]

7.8 Assessment of the Goodness of the Test Suite

GUIDE (delete before submission)

Coverage (statement/branch/requirement), effectiveness (defects found, mutation score), and limitations of your test suite.

[TODO: Assess the test suite.]

Chapter 8

Deployment and Go-Live

8.1 Deployment Environment

GUIDE (delete before submission)

Server/cloud platform, OS, configuration, containers, domain/SSL, environment variables (never print secrets!). Include a deployment diagram.

[TODO: Describe the deployment environment.]

8.2 Deployment Process and CI/CD Implementation

GUIDE (delete before submission)

Step-by-step deployment process and how the pipeline of Section 4.5 was implemented (show real workflow output/screenshots).

1. **[TODO: Step 1]**
2. **[TODO: Step 2]**

8.3 Go-Live Strategy and Post-Deployment Monitoring

GUIDE (delete before submission)

Rollout plan (big-bang, phased, canary, blue–green), user training, support/helpdesk, monitoring tools (logs, uptime, error tracking, AI-model drift), rollback plan.

[TODO: Describe go-live and monitoring.]

Chapter 9

Sustainability Plan

9.1 Scalability

GUIDE (delete before submission)

How the system handles growth in users/data: horizontal/vertical scaling, caching, load balancing, DB sharding, queueing, scaling of AI inference.

[TODO: Describe scalability.]

9.2 Flexibility / Customization

GUIDE (delete before submission)

Features that allow future modification: configuration files, plug-in architecture, modular design, theming, multilingual support.

[TODO: Describe flexibility and customisation.]

9.3 Maintainability and Future Enhancements

GUIDE (delete before submission)

Maintenance plan (bug fixing, updates, dependency upgrades, model retraining, documentation) and a list of proposed future features.

[TODO: Describe maintainability and future work.]

Acknowledgement

GUIDE (delete before submission)

Thank course teachers, supervisors, mentors, organisations, or users who helped (about 1 paragraph).

[TODO: Write acknowledgement.]

Bibliography

[1] R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach*. McGraw-Hill, 9th ed., 2019.

[2] I. Sommerville, *Software Engineering*. Pearson, 10th ed., 2016.

[3] OWASP Foundation, "Owasp top 10: 2021," 2021.

Appendix A

Full Source Code Snapshots & Repository Link

GUIDE (delete before submission)

Give the repository URL, commit hash / release tag submitted, and key source files. Use `\lstinputlisting` to include real files (keep it to the important ones).

Repository: **[TODO: paste GitHub/GitLab URL here]** Commit/tag: **[TODO: hash]**

Appendix B

Complete Test Cases (Detailed Tables)

GUIDE (delete before submission)

Place ALL test cases here (one longtable per module). Chapter 7 holds only the summary/sample.

Table B.1: Complete test cases

ID	Scenario	Steps	Data	Expected	Actual	Status
TC-01	[TODO: ...]	[TODO: ...]	[TODO: ...]	[TODO: ...]	[TODO: ...]	[TODO: ...]

Appendix C

Sample Screenshots of the Working System

GUIDE (delete before submission)

Screenshots of the final running system with one-line captions (put files in `images/`).

[Figure placeholder: `screenshot_home.png`]

Figure C.1: Home screen

Appendix D

Meeting Minutes / Progress Logs

GUIDE (delete before submission)

Record date, attendees, agenda, decisions and action items for each meeting (at least one per week).

Table D.1: Meeting log

Date	Attendees	Decisions	Action items
[TODO: dd-mm-yy]	All	[TODO: ...]	[TODO: ...]

Appendix E

Additional Diagrams

GUIDE (delete before submission)

Deployment diagram, component diagram, state diagrams, and any other diagrams that did not fit in the main chapters.

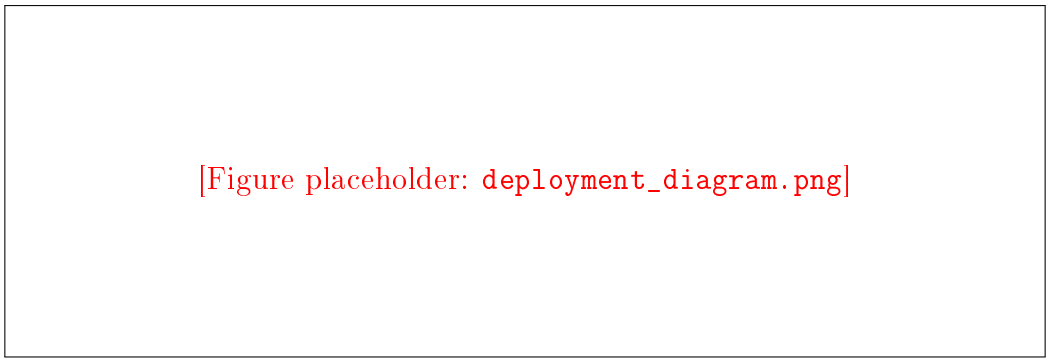


Figure E.1: Deployment diagram

Appendix F

AI Model Documentation / Training Details

GUIDE (delete before submission)

(If applicable) Model card: intended use, dataset description, training hyper-parameters, hardware, training curves, evaluation, limitations, ethical considerations.

[TODO: Add model documentation or write "Not applicable".]